

Computer Vision:

Week-2 (26-30 Jan) Theory Lectures

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Keypoint Detectors: Harris and FAST

- A **keypoint detector** is an algorithm used to find important and distinctive points in an image.

These points are called: keypoints / interest points / feature points

Keypoints usually correspond to:

- ✓ **Corners**
- ✓ **Junctions**
- ✓ **High texture regions**

Why Keypoints are Important?

-Keypoints should be:

- Repeatable (detected again in another image)
- Distinctive (easy to differentiate)
- Robust to noise and illumination changes

Among all image regions:

Flat regions → not useful

Edges → ambiguous

Corners → best keypoints

1. Harris Corner Detector

A corner is a point where image intensity changes strongly in both **horizontal and vertical directions**.

Harris detector identifies such points.

Conceptual Explanation

Imagine placing a small window on an image and shifting it slightly:

<i>Image Region</i>	<i>Intensity Change</i>
<i>Flat region</i>	Very small
<i>Edge</i>	Large in one direction
<i>Corner</i>	Large in all directions

- *Corners show high variation in every direction.*

How Harris Detects Corners

I. Compute image gradients:

- Horizontal change I_x*
- Vertical change I_y*

II. Analyze how intensity changes when the window moves.

III. Form a structure matrix using gradients.

IV. Compute a corner response value.

Corner Decision

Based on intensity variation:

- **Flat area** → no keypoint
- **Edge** → rejected
- **Corner** → selected as keypoint

Characteristics of Harris Detector

Advantages

- Very accurate corner detection
- Rotation invariant
- Strong mathematical basis

Limitations

- *Not scale invariant*
- *Computationally slow*
- *Sensitive to noise*

2. FAST Detector

(Features from Accelerated Segment Test)

*It is a corner detection algorithm
used in computer vision.*

FAST is used to:

- quickly find corner points in an image.*
- It is designed mainly for:*
- Real-time video*
- Object tracking*
- Robotics*

2. FAST Detector

(Features from Accelerated Segment Test)

FAST Corner Test

A pixel P is considered a **corner** if:

$$| I(x_i) - I(p) | > t$$

for **at least N continuous pixels** on the circle.

The surrounding pixel must be:

- **much brighter**, or
- **much darker**

than the center pixel.

Let:

- P_x = center pixel
- I_p = intensity of center pixel
- X_i = surrounding pixels on the circle
- I_{xi} = intensity of surrounding pixel
- t = threshold value

- **Brighter" condition** used in computer vision algorithms to determine if a specific neighboring pixel is significantly more intense than a reference pixel. It is a core component of the **FAST (Features from Accelerated Segment Test)** corner detection algorithm.

$$I(x_i) \geq I(p) + t$$

- **Darker" condition**, the logical opposite of the "Brighter" condition we just discussed. In the context of computer vision and the FAST (Features from Accelerated Segment Test) algorithm, it is used to identify pixels that are significantly dimmer than their surroundings.

$$I_{x_i} \leq I_p - t$$

Harris vs FAST (Conceptual Comparison)

FEATURE	HARRIS	FAST
Type	Corner-based	Intensity-based
Speed	Slow	Very fast
Accuracy	High	Moderate
Real-Time Use	no	yes
Scale Invariant	no	yes
Common Usage	Image matching	Tracking

Conceptual Setup:

You have a 3×3 grayscale patch:

100	100	100
100	150	100
100	100	100

Is the center pixel a **corner**, **edge**, or **flat**?

Conceptual Setup:

You have a 3×3 grayscale patch:

- Look at intensity change in **x-direction**:
- Left/right differences: $100 \rightarrow 150 \rightarrow 100$ large change $\rightarrow$ gradient exists
- Look at intensity change in **y-direction**:
- Up/down differences: $100 \rightarrow 150 \rightarrow 100 \rightarrow$ large change $\rightarrow$ gradient exists
- Both x and y gradients are strong $\rightarrow$ **corner** detected

*Center pixel is a **corner**.*